NASA Kepler Data + AstroNet-style 1D CNN

Machine learning pipeline for classifying stellar transit signals as confirmed planets or false positives

Team: Francisco medina | **Tier:** 2

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The Problem

What Real-World Problem

Astronomers need to distinguish genuine exoplanet transit signals from false positives in Kepler light curve data

Who Cares

Astrophysicists, NASA researchers, and exoplanet discovery teams analyzing Kepler mission data

Why Important

Manual classification is time-consuming and subjective; automated systems enable faster discovery of potentially habitable worlds

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My Solution

Automated exoplanet detection system using deep learning to classify Kepler light curves

1

Input

Kepler star ID, orbital period, transit epoch

2

Processing

1D CNN analyzes phase-folded light curves

3

Output

Planet detection probability (0-100%)

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Technical Approach

CV Technique

Binary classification of time series data

Model

1D Convolutional Neural Network (AstroNet-style architecture)

Framework

TensorFlow/Keras with scikit-learn for evaluation

Why This Approach

1D CNNs excel at detecting patterns in sequential data like light curves. The architecture mimics proven AstroNet designs that achieved high accuracy on NASA exoplanet data.

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Data Plan

Source

NASA Exoplanet Archive — Kepler Cumulative KOI Catalog (public dataset)

Size

Up to 400 light curves attempted (200 confirmed + 200 false positives) — actual usable count depends on quality filtering

Labels

Pre-labeled by NASA: CONFIRMED (class 1) vs FALSE POSITIVE (class 0)

Quality Filters

4-stage pipeline: download success, NaN rate <10%, signal variance check, data sanitization

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System Diagram

01

Download KOI Catalog

Fetch labeled dataset from NASA archive

02

Retrieve Light Curves

Use lightkurve to download Kepler data

03

Preprocess Data

Flatten, fold, bin, normalize flux arrays

04

Train 1D CNN

Convolutional blocks + dense classifier

05

Evaluate Model

Confusion matrix, ROC, precision-recall curves

06

Predict New Signals

Classify any Kepler star by parameters

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Success Metrics

59.38%

Target Accuracy

Test Accuracy — Final model performance

0.573

ROC Performance

Test AUC — ROC curve performance

7.4ms

Processing Time

Processing Time — Per light curve prediction

Primary Metric: Test accuracy (59.38%) and AUC (0.573)

Secondary Metrics: Precision (59.57%), Recall (80.00%), processing time per prediction

Target vs Actual: Accuracy target was >80%, actual test accuracy was 59.38%. AUC target was >0.85, actual test AUC was 0.573. The model achieved strong recall (80%) but lower precision (59.57%), indicating it correctly identifies most planets but has false positives.

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Week-by-Week Plan

1

Week 1 (Feb 17)

Get dataset, set up environment, install dependencies

2

Week 2 (Feb 24)

Train/fine-tune 1D CNN model architecture

3

Week 3 (Mar 3)

Test, evaluate results, and complete documentation

4

Completed (Mar 6)

Project finished! 🎉

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Challenges & Backup Plans

Challenge: Data Quality Issues

Problem: Light curves with too many NaN values or poor signal-to-noise ratio

Plan B: Increase sample size from NASA catalog, use more aggressive filtering, or apply data imputation techniques

Challenge: Low Accuracy

Problem: Model fails to achieve target performance metrics

Plan B: Try deeper CNN architecture, add dropout layers, use data augmentation, or experiment with different hyperparameters

Challenge: Download Failures

Problem: NASA MAST archive unavailable or rate limiting

Plan B: Use cached dataset from previous runs, reduce sample size, or switch to alternative data sources

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Implementation Details

Compute Environment

- Google Colab (T4 GPU recommended)
- TensorFlow 2.x + Keras
- lightkurve library for data access
- scikit-learn for evaluation

Model Architecture

- 4 convolutional blocks (16-32-64-128 filters)
- MaxPooling between blocks
- Dense head with dropout
- Sigmoid output for binary classification
- Train/Val/Test split: 68% / 12% / 20%

Training Configuration

- Adam optimizer (1e-4 learning rate)
- Binary crossentropy loss
- 60 epochs with early stopping
- Batch size: 32
- Input shape: (201, 1)

Cost

Free: Google Colab runtime with GPU support

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